

Thermoluminescence dating

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Thermoluminescence dating

is an absolute dating technique that determines the age of object based on the strength of luminence produced

Principles

- ① An object when heated at high temperature absorbs some amount of energy.
- ② Using TL techniques this energy can be estimated based on the light produced by the object.
when heated again

Advantages

- ① Allow accurate estimation of age with limited error window
- ② Useful for dating cultural material

like pottery.

- ③ Even a small piece of material is enough to produce accurate results.

Use

- ① Determine the age of iron age cultures in Europe & west Asia
- ② Study of Mesoamerican cultures.

Limitations

- ① Still under process of development & the phenomenon is not completely understood.
- ② Not possible to date organic or inflammable material like bone, cloth, etc.
- ③ Require greater expertise to produce accurate result.

Thus, the dating has proven to be a crucial tool to understand timeline of human development.